

Lab W2D1

Question 1. Design and Analysis of the algorithms

(a) Consider an array of “Wooden blocks toys”. One of the features of these toys is that they are painted either Blue or Red. Devise an algorithm to keep all the Blue toys together at one end of the array and all Red toys together at the other end of the array. Is your algorithm in place? If not, what is the space complexity? What is the time complexity?

(b) Solve it for three different colors: Blue, Red and Green. . Is your algorithm in place? If not, what is the space complexity? What is the time complexity? Remember we are more concerned about the time complexity.

(c) Solve it for four different colors: Blue, Red, Green and Yellow. Is your algorithm in place? If not, what is the space complexity? What is the time complexity? Remember we are more concerned about the time complexity.

Question 2. Illustrate Quick sort.

Please use median of three rule.

(a) {1, 2, 3, 4, 5, 6, 7, 8, 9}

(b) {8, 7, 6, 5, 4, 3, 2, 1, 9}

(c) {9, 1, 8, 2, 7, 3, 6, 4, 5}

Question 3. Illustrate quickSelect.

Please use median of three rule.

(a) {1, 2, 3, 4, 5, 6, 7, 8, 9}, $k = 4$

(b) {8, 7, 6, 5, 4, 3, 2, 1, 9}, $k = 5$

(c) {9, 1, 8, 2, 7, 3, 6, 4, 5}, $k = 6$